

SI 1. Summary of the 3 reference dataset

Summary of the three raw datasets used as reference. These studies have provided in their SI, or after contacting authors, their raw particle data, meaning size information on every MP detected.

	Lu et al., 2022	Perera et al., 2022	Ueda et al., 2025
Medium	Wetlands	Indoor and Outdoor Air	Surface coastal waters
Location	Gold Coast, Australia	Sri Lanka	Tokyo bay, Japan
Sampling method	10L glass bottle, 25 μm stainless-steel filter	Stainless-steel air pump with 1 μm stainless-steel filter	Pump system with plankton net filtration (10 μm) and neuston net tow (350 μm)
Analysis method	FTIR after visual identification under stereomicroscope	μ -FTIR after visual identification under stereomicroscope	FTIR (manually picked for net sample)
Quality control	Procedural Blanks	Fields Blank	Procedural blanks
Recovery	90%	no data	96 % for pump and 98% for net tow
#MP counted	3191	104	pump: 3740 net: 537
MP size range reported	25 - 4946 μm	49 - 4831 μm	21 - 4960 μm

SI 2. Test on Virkar et Clauset (2014) MLE method for binned data.

Published MP binned PSD examples tested by the Virkar et Clauset (2014) MLE method. 'x' is the log of the geometric mean bin size, and $\Pr(X \geq x)$ (y-axis) is the log of the MP cumulative probability density distribution, which corresponds to the cumulative concentrations. Alpha_0 represents the negative of slope parameter a; bmin_0 is the lower LOD identified by the MLE algorithm; L represents the negative of intercept log (b). The purpose of illustrating MLE fitting of binned MP datasets, is that two out of eight fitted slopes are highly unrealistic (too steep) and do not permit MP dataset alignment.

